TEST SUGGESTED SOLUTIONS

(1) MEAN $\frac{\sum x_i}{n}$

$AM_A = 0.066$ (1) $AM_B = 0.016$ (1)

A IS DESIRABLE (1) AS IT HAS A HIGHER MEAN.

(b) STANDARD DEVIATION

$$\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

$Var_A = 0.01463$ (1) OR Var_A 0.011704.
$Var_B = 0.00403$ (1) OR Var_B 0.003224.

$STD_A = 0.120956$ (1) or 12.09% OR SD_A 0.108185 or 10.82%
$ST_B = 0.063482$ (1) or 6.35% | SD_B 0.05678 or 5.68%

B WOULD BE DESIRABLE (1), AS THIS HAS A LOWER STANDARD DEVIATION.

(c). $C.V. = \frac{E(R)}{\sigma}$

$CV_A = \frac{0.066}{0.12095}$ OR $CV_A = \frac{0.066}{0.108185}$

$CV_A = 0.54566$ (1) $= 0.610066$.

$CV_B = \frac{0.016}{0.063482}$ OR $CV_B = \frac{0.016}{0.05678}$

$= 0.25204$ (1) $= 0.281788$.

$\Rightarrow$ A IS DESIRABLE, AS IT HAS HIGHER RETURN PER UNIT OF RISK (1)

(d) IF THE TWO HAVE THE SAME EXPECTED RATE OF RETURN. 1

(e). IF THE TWO HAVE MAJOR DIFFERENCES IN EXPECTED RATE OF RETURN AND STANDARD DEVIATION (OR VARIANCE) 1

② $u(w) = -e^{-\beta w}$ $w > 0$, $\beta > 0$.

(a) $u'(w) = \beta e^{-\beta w}$ 1/2 $u''(w) = -\beta^2 e^{-\beta w}$ 1/2

$$ARA = -\frac{u''(w)}{u'(w)} = -\frac{(-\beta^2 e^{-\beta w})}{\beta e^{-\beta w}}$$ 1/2

$$= \beta$$

CONSTANT ARA (CARA) 1/2

RRA = ARA × w = βw 1/2

INCREASING RRA 1/2

(b)(i) AMOUNT INVESTED IN RISKY ASSET DOES NOT CHANGE AS WEALTH INCREASES. — (CARA) 2

(ii) PROPORTION INVESTED IN RISKY ASSETS DECREASES AS WEALTH DECREASES. (IRRA). 2

3. (a). (i) PREFER MORE TO LESS. 1

(ii) PREFER MORE TO LESS AND BE RISK AVERSE. 1 1

(b). MOST STUDENTS PROVIDED THE FOLLOWING ~~THEREFORE~~ I HAD MARK THIS.

$E(P_1) = 6.75$ 1/2 $\delta_1^2 = 15.22$ 1/2 $E(P_2) = 5.37$ 1/2

$\delta_2^2 = 4.25$ 1/2 $\Rightarrow$ $CV_1 = 0.4435$ 1/2, $CV_2 = 1.2635$.

$\Rightarrow$ INVESTMENT 2 WILL BE CHOSEN AS IT HAS A HIGHER RETURN PER UNIT OF RISK 1

3.C
(a)

P	F(1)	F(2)	F(1) − F(2)	Σ(F1 − F2)
1	0.25	0	0.25	0.25
2	0.25	0	0.25	0.5
3	0.25	0.33	−0.08	0.42
4	0.25	0.33	−0.08	0.34
5	0.25	0.66	−0.41	−0.7
6	0.25	0.66	−0.41	−0.48
7	0.75	0.66	0.09	−0.39
8	0.75	1	−0.23	−0.64
9	0.75	1	−0.25	−0.89
10	0.75	1	−0.25	−1.14
11	0.75	1	−0.25	−1.39
12	1	1	0	−1.39

THE IS NO SOSD, THE SIGN of $\Sigma(f_1 - f_2)$ IS NOT MONOTONE, i.e. IT CHANGES

4 (a). S HAS A HIGHER RETURN AND LOWER VARIANCE SO IT IS PREFERRED IN THE MEAN-VARIANCE FRAMEWORK.

(b) RELAX ASSUMPTION THAT INVESTORS CHOOSE PORTFOLIOS SOLELY ON THE BASIS OF EXPECTED RETURN AND VARIANCE OF RETURNS.

(c). $E(P) = 0.75 \times 10 + 0.25(8)$

$= 9.5\%$

(d) $\sigma_p^2 = (0.75)^2(16) + (0.25)^2(25) + 2(0.75)(0.25)\sqrt{16}\sqrt{25}\times 0.3$

$= 12.8125\%$

$\sigma_p = 3.57945$

(d). AMOUNT INVESTED IN S, w_s WILL BE

$$w_s = \frac{Var(m) - Cov(S,m)}{Var(S) + Var(m) - 2Cov(S,m)}$$

$$= \frac{25 - 4\times5\times0.3}{16 + 25 - 2\times4\times5}$$

$$= \underline{\underline{0.655173}}$$

$\therefore$ 0.344828 INVESTED IN m

(e). # THE STUDY SUGGESTS THE CORRELATION BETWEEN S AND m WILL INCREASE.

THEREFORE A PORTFOLIO WITH POSITIVE WEIGHTS OF M AND S WILL HAVE HIGHER VARIANCE.

IF CORRELATION INCREASES, MINIMUM VARIANCE PORTFOLIO WILL CONTAIN RELATIVELY HIGHER AMOUNTS OF S AND RELATIVELY LOWER AMOUNTS OF M.

Question 5: $\longrightarrow$ <u>IGNORE</u>.

6

(a) MARKET PORTFOLIO

MARKET CAP 30,000,000 20,000,000

w_i = 0.3 0.7 (1)

(b) mean returns.

Asset 1 = 21.8% (1/2)

Asset 2 = 14.9% (1/2)

$\Rightarrow E(R_m) = \sum w_i E(R_i)$

$= \underline{16.97}$ (1)

STD. DEV$(R_m) = \delta m$

$= \sqrt{E\left(\sum w_i R_i\right)^2 - (E(R_m))^2}$

$= \underline{3.57\%}$ (2)

MARKET PRICE OF RISK $= \dfrac{E(R_m) - rf}{\delta m}$

$= \dfrac{0.1697 - 0.05}{0.0357}$

$= \underline{3.35}$ (1)

(c). FROM SML $\beta = \dfrac{E(R_i) - Rf}{E(R_m) - Rf}$

$\Rightarrow \beta_1 = \dfrac{21.8 - 5}{16.97 - 5} = \underline{1.403509}$ (1) or $\underline{1.40}$

$\beta_2 = \underline{0.827068}$ (1) or $\underline{0.83}$

(d) BECAUSE THE BETAS ABOVE ARE OBTAINED VIA CAPM EQUILIBRIUM CONDITION, BOTH STOCKS ARE PRICED AT EQUILIBRIUM (1) i.e. CORRECTLY VALUED $\longrightarrow$ ONE CAN CONFIRM THIS BY ALPHAS (= ZERO) OR PLOT SML $\Rightarrow$ ALL ON LINE.

IF EXPECT MARKET TO BE 1 BULLISH ~~Het~~ TAKE LONG POSITION ON 1 ASSET I

IF EXPECT MARKET TO BE 1 BEARISH TAKE LONG POSITION ON 1 ASSET 2.

7 (a) MULTI-FACTOR IS MORE ROBUST AND INTUITIVELY APPEALING, SINCE IT USES MORE FACTORS (INCLUDING MARKET PORTFOLIO OF CAPM), THUS 1 MAY BETTER EXPLAIN RETURNS.

\# IT ALLOWS FITTING ~~SELECTION~~ OF STOCKS DATA ~~ALSING~~ BASED ON SPECIFIC INDEP-SENSITIVITY CRITERIA → 1 THUS MAY PREDICT FUTURE BETTER AS IT USES MORE DATA.

(b). $\alpha_p = -0.5(0.03) + 1.5(0.05) + 0(0.1)$

$= 0.06.$ 1

$\beta_{p1} = -0.5(1) + 1.5(3) + 0(1.5)$

$= 4.$ 1

$\beta_{p2} = -0.5(-2) + 1.5(2) + 0(1.5).$

$= 5.$ 1

$R_p = 0.06 + 4I_1 + 5I_2 + \varepsilon_p.$ 1

(c). $w_A + w_B + w_C = 1.$

$w_A + 3w_B + 1.5w_C = 1$

$-4w_A + 2w_B + 1.5w_C = 0.$

SUBSTITUTION GIVES.

$w_A = 0.25$ 1 $w_B = -0.25$ 1 $w_C = 1.$ 1

\# CALCULATE $\alpha_p = 0.095$ $R_p = 0.095 + I_1 + \varepsilon_p.$

⇒ COMMENT – THE 1 ABOVE IS A PURE FACTOR PORTFOLIO.